

An Electronic Travel Aid Device to Help Blind People Playing Sport

Sylvain Ferrand, François Alouges, and Matthieu Aussal

Blind people can locate and track a sound source with remarkable efficiency. Following this principle, we have developed an autonomous embedded device capable of guiding people using a spatialized virtual sound source. In this device, we use both the information provided by an accurate and low-latency real-time location system (RTLS) and the head orientation data provided by a low-cost inertial unit (IMU) to produce in real-time spatialized sound by binaural filtering. The system described in this paper has been designed to minimize the overall latency for both the location system and the binaural audio rendering engine. It demonstrates that it is possible to guide people by tracking spatial sound accurately, naturally, and with minimal training.

Introduction to Electronic Travel Aid Devices

Visual impairment is a prevalent disability that strongly affects people's autonomy. Indeed, 216 million people are visually impaired in the world while 36 million people are completely blind [1]. Conventional aids, such as white canes or guide dogs, help these people have more autonomy in their movements. However, these traditional means remain limited. Indeed, while the white cane gives information confined to the immediate environment, the dog can perform more complex tasks but still with very limited navigation planning capabilities. Electronic Travel Aid (ETA) devices have thus been developed to overcome these shortcomings.

ETA devices already have a long history. At the end of the 19th century, the Polish researcher Kazimierz Noiszewski built a device [2] using a photoelectric cell that allowed blind people to distinguish light from darkness by varying the intensity of a sound. In 1912, d'Albe built the *exploring Optophone* [2], working on the same principle. Unfortunately, providing this kind of information to blind people is of little use, and much later, some systems have tried to convert complete images into tactile [3] or audio stimuli [2], [4].

Most of these projects can be classified among the Sensory Substitution Devices (SSD), a concept that was introduced 50 years ago by Bach-y-Rita [3]. The general idea is to transmit

data through a different sensory channel than the one usually used, for example, by replacing visual information with sound encoding.

In general, different approaches to SSD can be distinguished (notice that some projects use several techniques simultaneously):

1. Direct scene representation: It was the first approach to SSD. It consists in trying to convert a whole visual scene into another sensory channel. The first device of this type [3] constituted electro-tactile actuators fixed on the back of a person giving a representation of an image filmed by a camera.
2. Environment analysis for scene description and obstacle avoidance: These devices sense the environment to extract useful information for the user. Some of them use laser sensors, for instance, the electronic white cane by Farcy *et al.* [5]. Others are based on computer vision technology as in Johnson and Higgins' obstacle avoidance device [6] which provides vibrotactile stimuli [6] or Bujacz *et al.* device [7] with sonification techniques.
3. Navigation devices focusing on guidance along a planned path: While devices in (1) try to provide a full representation of the environment by sensory substitution and (2) give more parsimonious pieces of information, helping to understand the local environment, the third method consists in giving only information about the direction to follow. This approach significantly reduces the amount of information to be conveyed and then facilitates the interaction with the user, improving the speed of information perception and reducing the cognitive load. This approach was first explored by Loomis [8] with a *Text to Speech* interface and later with binaural stimuli [9].

Even if many SSD devices for environment representation have been developed in the past 50 years (see [10] for a complete review on tactile SSD), very few devices are used in practice by blind people. Major disadvantages of these techniques are: they are uncomfortable and have pervasive aspects

(blocking ears, weight, etc.), training requirements, and the cognitive load that they involve.

In this paper, we focus on the guidance task (case 3) and its effective implementation. For this purpose, we do not try to encode environmental information except the direction to follow. Such guiding methods have been already experimented in different research projects, for example, GpsTunes [11], On-track [12], Argus [13], Navig [14] and Swan [15] that are mostly based on GPS positioning systems for the localization.

Furthermore, visually impaired people can localize and follow sound sources with remarkable accuracy. They often use this ability in everyday activities, for example, to follow someone by hearing his/her voice or footsteps. In principle, it is possible to use this faculty to guide blind people with spatialized sound generated in real-time and heard with simple headphones (binaural listening). Indeed, if we can localize the person accurately in real-time, we can create a virtual sound source at a certain distance ahead of him/her and guide him/her to follow a pre-defined path.

Due to the nature of binaural stimuli, the user can analyze the information very quickly, and we thus need to provide him/her with fast, real-time feedback. For this purpose, the whole device is designed to minimize the latency, including localization and sound rendering.

Device Design

We aim to provide the user with a natural sound source located in front of him/her that shows the direction to follow. In order to realize that goal, we need to locate the user as accurately as possible and create a spatial sound which, being rendered on headphones, has to be furthermore adapted, depending on the orientation of his/her head. Moreover, the device must be portable and work in real-time.

Indeed, the main idea behind a binaural guidance device is to allow the person to follow the sound source as naturally as possible. To do this, it is essential to give the user a realistic feeling that is consistent with what would be perceived in a real situation. For a realistic experience, each user's movement must be taken into account and the audio rendering modified accordingly.

The complete device, which is sketched in Fig. 1, includes therefore a head-tracker, a real-time localization system, and a processing unit. We describe hereafter its components and both the material and software architectures.

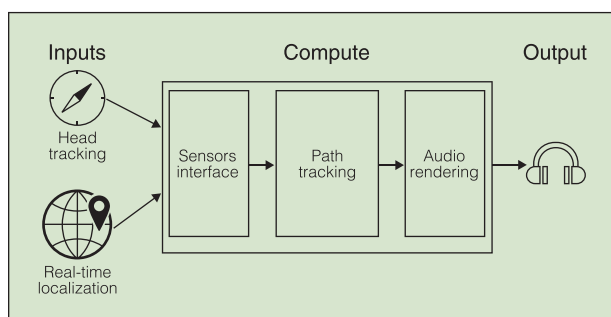


Fig. 1. General device overview.

Head Tracking

The head-tracker must measure the orientation of the head of the subject. It consists of a low-cost Microelectromechanical assembly of three sensors (MEMS) that measures simultaneously, in the three orthogonal axes, the acceleration, the rotational velocity, and the earth's magnetic field. Unfortunately, these sensors do not provide any direct orientation information, and the preceding measurements must be fused together with appropriate algorithms. Typical data fusion methods are:

- A combination of high-pass and low-pass filters [16] (often called *Complementary filters*);
- Kalman filters [17];
- Specialized algorithms such as Mahoni or Magdwick's algorithms [18]. The Magdwick's filter is a complementary filter that uses a gradient descent algorithm and is based on a quaternion representation. This algorithm offers a precision of estimation comparable to Kalman filters but with less computation. That is why it is widely used for IMU data fusion.

We have experimented with various devices: some all-in-one chips combining MEMS sensors and data fusion units and some home designed devices with MEMS and microcontrollers running data fusion algorithms. A comparison is given in Table 1.

The devices that have been tested show similar accuracy but with significantly different refresh rates and magnetometer calibration processes. In our system, we chose a head-tracker built with the low-cost TDK-Inversense MPU9250 MEMS sensor associated to an ESP32 microcontroller with built-in USB and Bluetooth interfaces. It can perform data fusion at 500 Hz with 0-drift using Madgwick's algorithm. This head-tracker can be connected to the main system through a USB or Bluetooth connection. Our implementation is derived from Kris Winer's work [19] which has developed several codes for data fusion with the arduino API and low-cost IMU.

Real-time Localization

In the literature [11]–[13], [15], several ETA devices use Global Navigation Satellite Systems (GNSS) to provide localization data for navigation. The accuracy of GNSS systems is typically in the range of 2–5 m horizontal with an update rate in the order of 1–10 Hz (see Table 2). Moreover, they are not usable indoors. This is not sufficient for our purpose because giving immediate feedback to the moving user (even for small or slow

Table 1 – Comparison of head tracking performances

Device	Heading accuracy	Refresh rate
Bosch BNO055	3–5°	100 Hz
Hillcrest BNO080	2–4°	400 Hz
Sparkfun Razor IMU	3–4°	15 Hz
MPU9250+Magdwick	3–4°	500 Hz

Table 2 – Comparison of Positioning Systems

Technology	Accuracy	Refresh rate	In/out	Cost
GNSS (GPS)	2–5 m	1–10 Hz	outdoor	low
GPS + IMU	2–5 m	100 Hz	outdoor	low
UWB	10 cm	10–15 Hz	in/out	medium
RTK-DGNSS	10 cm	1–10 Hz	outdoor	high

moves) is crucial. Therefore, we have experimented with various commercial systems, and we also propose methods for indoor localization.

Outdoor Localization: Differential GPS (DGPS) is an improvement of GPS systems. It uses some reference stations that transmit the difference between the positions indicated by the satellites and their known actual positions. The DGPS principle is based on the fact that at neighboring positions, measurement errors are very similar and the knowledge of the exact position of the reference stations can be used to correct the measurements of the mobile receiver. This system makes it possible to improve location accuracy to about 10 cm under the best conditions of implementation and observation.

The cost of commercial receivers is currently falling sharply, which may make this technology, which has been very expensive until now, usable for consumer applications. It is also possible to develop a DGPS system using the adapted hardware of open-source software components [20]. In our device we use the Ublock Neo C94-M8P D-GNSS module (GPS/GLONASS) with a small portable patch antenna that can be fixed on the shoulder or the helmet of the user. Our experiments show that the typical precision of the system is typically about 10–20 cm which is satisfactory for our application. Conversely, a conventional (non-differential) GPS would not be accurate enough.

Indoor Localization: Indoor movements are also essential for blind users, for example, to navigate inside a building or playing sport in a gym. Indoor localization is a very active field of research. A wide range of methods have been proposed, using proximity to beacons, triangulation, fingerprinting of the environment or dead reckoning. They are often based on technologies such as Wi-Fi [21], Bluetooth [22], or Ultra-Wideband (UWB) [23]. Most indoor localization systems use the *Received Signal Strength* (RSS) of a radio signal. This technique consists in estimating distances, considering that the power of a received signal decreases with the covered distance. Unfortunately, the decay rate is difficult to model in a non-free-field environment. Besides, these technologies are sensitive to shadowing or antenna orientation problems (polarization).

On the other hand, technologies using the Time of Flight (ToF), i.e., the time traveled by the signal between the transmitter and the receiver, are usually much more accurate and robust. Measuring precise timing for a radio wave has been a challenge for embedded electronics until recently, but these technologies are now available. We experimented with UWB,

a technology that allows a fast and precise distance measurement by using the Time of Flight of a radio signal. In association with the *Two Way Ranging protocol* (TWR) [24], it is possible to measure the distance between a master UWB transceiver (the Tag) and multiple slaves (the Anchors). Using this protocol, we

measure the round-trip time of a data frame sent from the Tag and replied by the Anchors. This elapsed time permits the distance between the Tag and the Anchors to be deduced. Based on ToF, this technology is very robust in multi-path environments and relatively insensitive to shading of the sensors and RF interference. It is also low-cost, low power, and gives high precision measurements. The UWB transceivers that we use are based on the DW1000 UWB chip from DecaWave.

Once the distances to all the reference beacons/anchors are known, the position of the receiver can be estimated by trilateration. Solving a trilateration problem consists in finding the intersection point of circles or spheres with noisy measurements. We have proposed a method in two steps:

- In an initialization step, we accumulate the measured distance to each Anchor while moving around them with the device. This enables us to estimate the position of both the Anchors and the Tag along the realized path using a Least Square Error method (it amounts to solving a system of quadratic equations);
- Once the position of the anchors is known, we can start the real-time localization. For this purpose, we use an Unscented Kalman Filter (UKF) which is known to be well suited to compensate for the Gaussian noise produced by UWB measures and the nonlinear measurement function. Interested readers are referred to [25] for a careful study and a complete description of the method.

This technique makes it possible to localize a mobile receiver 10 times per second with an accuracy of 10 cm within a 30 m x 30 m square.

Audio Rendering: As already explained, the interaction with the user is done using binaural sound [26]. This technique seeks to reproduce the properties of natural listening by digital filtering and allows the user to perceive virtual sound sources in space, using conventional headphones. Indeed, the human auditory system uses several cues for sound source localization. These cues include the difference in arrival times of the sound between the ears (interaural time differences, ITD), the differences in sound level entering the ears (interaural level differences, ILD), and spectral information (direction selective filter mostly related to the geometry of the pinna).

The filter responses that characterize how ears receive a sound from a point in space are called HRTF (for Head Related Transfer Functions), and they can be used to synthesize a binaural sound that seems to come from specific locations. It has been shown [27] that HRTF filtering allows good accuracy for estimating the azimuthal angle of a sound source, a

Table 3 – Computing time for audio spatialization on various platforms

	1 source	6 sources	24 sources
Mac (i5, 2.4 GHz)	0.9 ms	2.5 ms	6.0 ms
Arm 8 (Rasp. Pi 3)	4 ms	11 ms	no
Arm (Odroid XU4)	3 ms	9 ms	no
Intel Edison (Atom)	9 ms	13 ms	no

precision that can be furthermore improved by taking into account the movements of the head [28] in real-time. Therefore, once coupled with a head-tracker such as the one described above, guiding someone with a spatialized binaural sound becomes possible.

The binaural sound engine is mostly implemented in Python. It also uses Cython for heavy processing tasks. This implementation is derived from the software initially developed in Matlab language by M. Aussal and F. Alouges [29], [30] based on the convolution of sound signals with HRTFs. The software can process multiple audio sources (i.e., it can spatialize multiple sound sources in parallel) and allows a selection of filters in real-time according to the relative position of the user and sources. We use a set of HRTFs derived from the subject 1040 of the IRCAM *Listen* database [31]. To give precise direction indications and to limit artifacts related to filter changes, HRTF measurements have been previously interpolated every 2°. In addition, we have reduced the processing buffer as much as possible in order to limit the latency.

This engine has been tested on PC/Mac, ARM processor, and Intel Edison. Table 3 shows the performances of different embedded systems. Considering a typical 20 ms audio buffer, small embedded platforms can spatialize easily 6 audio channels (sound sources) simultaneously.

For only one source, which corresponds to our typical use, the audio latency is as low as a few ms on the hardware platform that we are using (Odroid XU4).

Hardware Architecture: To be easily transportable, the device is built around a compact, low-power ARM platform. On this platform, the different sensors are connected via the USB bus. The complete platform (including batteries) weighs less than one kilogram with an autonomy of about 2 hours.

- The motherboard: Since the processor has to perform simultaneously both audio processing and localization tasks in real time, we use a modern ARM-based high-performance single board computer. The modularity of the system's software design allows the effective use of multi-core system and the big.LITTLE architecture of the ARM v7 processors appears to be well suited for our purpose. The audio engine requires to store a large number of HRTF filters (remember that we use a pair of filters every 2° covering a complete sphere). For performance reasons, all these filters are preloaded in memory at startup, which, depending on the size of the filters, can represent up to 10 MB of storage. This disqualifies a

large number of signal processing processors (DSP) and we have therefore chosen to use general-purpose ARM processors. A typical choice is the Odroid XU4 card, with octa-core Exynos 5422 big.LITTLE processor, but Raspberry PI 4b cards and other low-cost development boards may also be suitable.

- The Power supply module: The battery has to provide power for all the modules, i.e., approximately 15 W in working condition. We are using a 3000 mAH 2 cells (7.4V) LiPo batteries connected to a 5 A max step-down 5 V converter. This setup permits to connect bigger LiPo batteries for longer use.
- Audio Output: As visually impaired people cannot be isolated from the outside, we use bone conduction headphones (model *AfterShokz Sportz titanium*) intended for sports practice. It has been already shown by Bujacz *et al.* [27] that it is possible to use bone headphones for sound localization, at the cost of a small loss of accuracy.

Software Architecture: We have tried to make the software modular while minimizing the latency associated with communications between the different software and hardware layers. It is based on a multi-tier architecture where each logical component exchanges data via a REST API [32]. The main program acts as a REST server and manages communications with clients. It also includes the communication interface with the sensors (head-tracker and positioning system) and the filtering system.

The binaural engine is a client that continuously obtains server location data via REST HTTP requests. The REST server acts as an abstraction layer for the sensors, so the binaural engine obtains location information independent of the location technology (GPS, RTLS systems, etc.) and whatever their refresh rate.

A second REST client provides path tracking, visualization, and data recording. It should be noted that with this HTTP-based REST architecture, clients can be hosted on a remote computer; especially, the visualization or analysis of the data could well be transferred to another machine.

Fig. 2 shows the message exchanges and the organization of the different processes around the Representational State Transfer (REST) server. By separating each task into different processes, the software is able to use multi-core embedded processors efficiently.

Use Cases and Results

The complete device can be seen in Fig. 3. It has been tested with normally-sighted and blind people walking or playing sports (running or roller skating) on large open fields, gyms, and athletic tracks.

The device was initially designed for sports practice (for running on an athletic track, for example) and in a controlled environment, since we do not provide any system for obstacle detection yet. The device is typically used in two steps:

- First, a trajectory is recorded by a normally-sighted person equipped with the device;

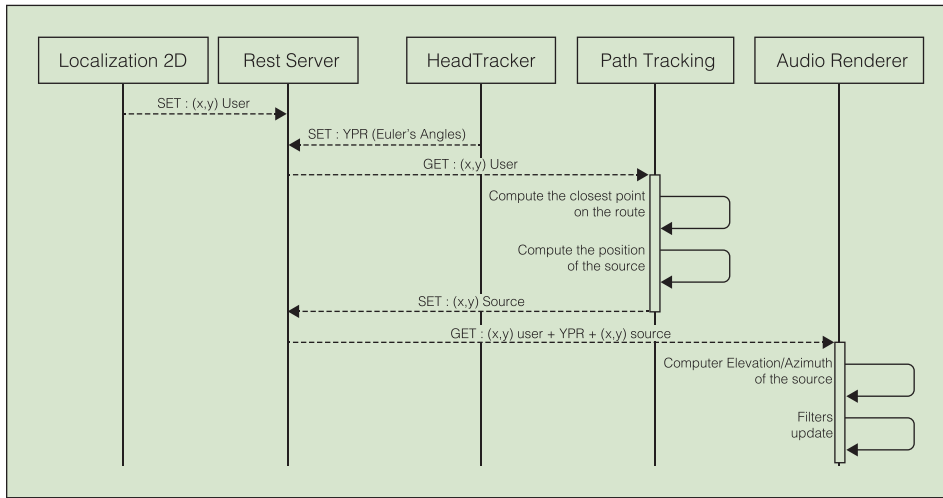


Fig. 2. Message exchanges between software modules.

► Then, a blind person wears the device and the guidance starts (i.e., by pressing the guidance button) at any location close enough to the recorded track. The device directs the user to the nearest point on the track and starts the guidance.

The device usually requires very little training, and the user can follow the sound almost immediately. Nevertheless, training allows the user to increase his/her performance and move faster. For testing purposes, we recorded a large elliptical track on a 30 m x 25 m gym. In the experiments described next, we measure, at any time, the user's position and the distance to the path to be followed, denoted by e_t . For m measurements, we compute the averaged error over a path:

$$e_{mean} = \frac{\sum |e_t|}{m} \quad (1)$$

Experiment 1

A blind user is equipped with the device. We ask him to walk slowly and let himself be guided by the system along an elliptical track. The results are given in Fig. 4 (walking, user 1). The user follows the track with very good accuracy: the average deviation from the ideal route is $e_{mean} \sim 17$ cm (max 53 cm, $\sigma = 0.12$). Such a precision can typically allow a narrow path to be followed. In this experiment, the average speed of 1.8 km/h corresponds to a slow walking.

Experiment 2

A blind roller-skater is equipped with the device. We ask him to let himself be guided by the system and try to gain speed, if possible. The results are shown in Fig. 5. The user has been able to reach an average speed of 8.3 km/h with peak speed at 11 km/h. In these conditions, the average error increases to 0.66 m (max 2.27 m, $\sigma = 0.45$), but this remains reasonable for practical use. The user describes the experience as quite comfortable.

Experiment 3

We believe that in the preceding experiment, the user did not have enough time to locate the sound accurately, which could explain the increase in error. To address this problem, we introduce a new type of feedback.

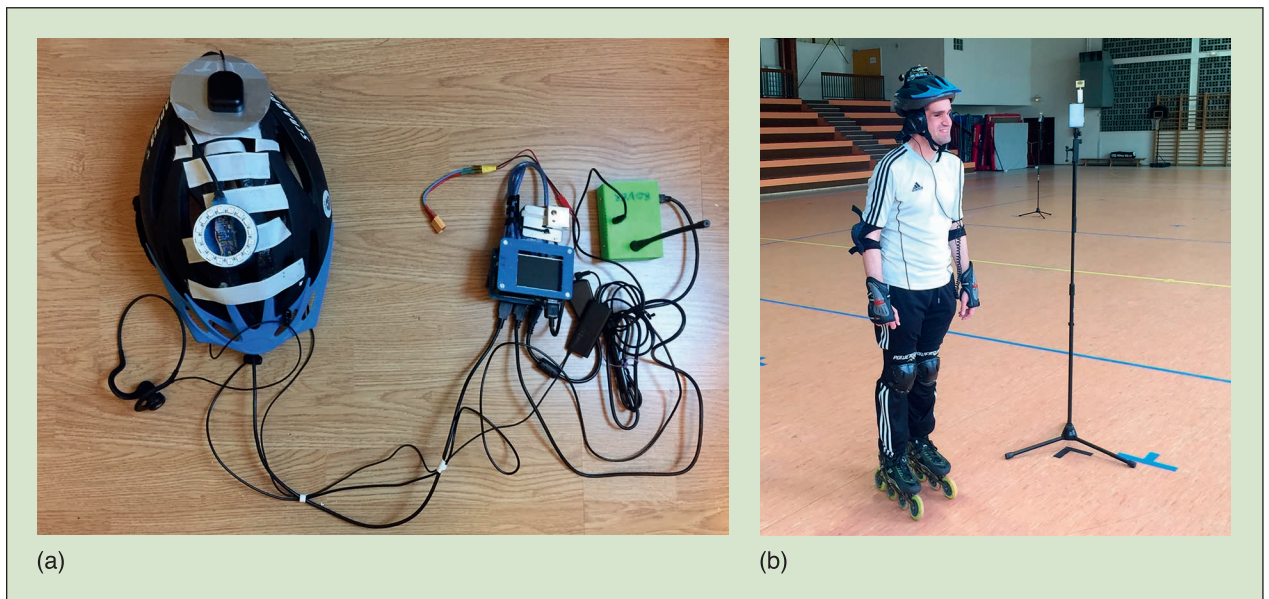


Fig. 3. The complete system. (a) The helmet with GNSS antenna and the head-tracker and , the processing unit in a blue case with the D-GNSS receiver in green (typically placed in a small backpack). (b) The device worn by a blind roller-skater. Two UWB antennas are also visible.

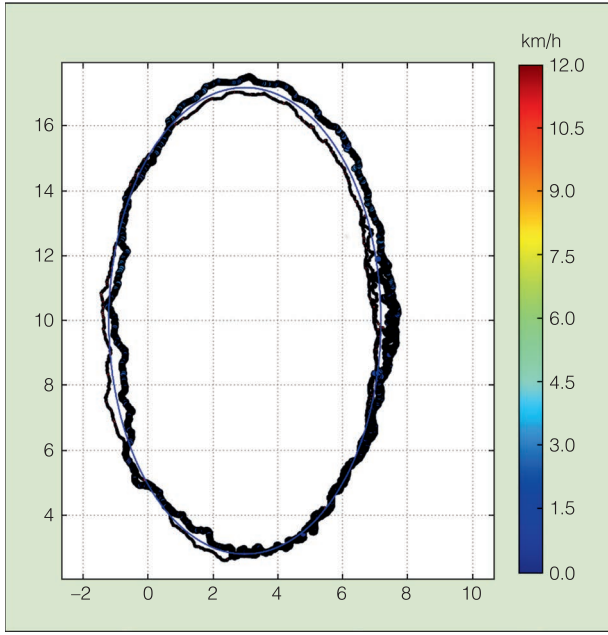


Fig. 4. A blind man walking along an elliptical path. The blue line is the course to follow and dots represent the user position with color for velocity (blind subject #1).

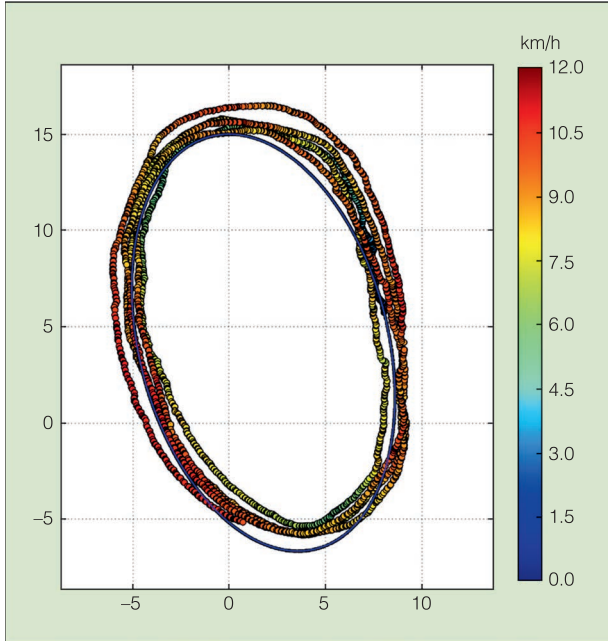


Fig. 5. Roller skating along an elliptical path (Blind subject #2).

Considering the positioning error e_t , we introduce a proportional feedback:

$$F_t = K_p e_t \quad (2)$$

and we move the source by F_t in the direction opposite to the error from the theoretical position on the path (Fig. 6). The parameter K_p is the proportional feedback coefficient and can be tuned empirically. In some cases, the user follows the path

without ever reaching it (because the accuracy of the perception of sound sources is limited). To address this problem, we can also add a term that takes into account the accumulation of the error over a time window (similar to a well-known proportional-integral feedback controller):

$$F_t = K_p e_t + K_i \sum_j e_j \quad (3)$$

We therefore carry out guidance tests using proportional-integral feedback, and we observe a significant increase in accuracy. Fig. 7 shows the path of user 2 under the same conditions as the test illustrated in Fig. 5 but with the feedback turned on.

For this experienced blind roller skater, with PI-feedback, the maximum speed was up to 12 km/h and the average error was 0.5 m (maximum 1.3 m, $\sigma = 0.43$). Table 4 summarizes the

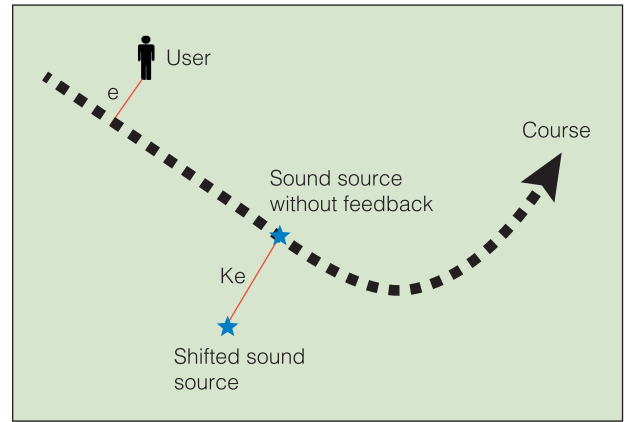


Fig. 6. Proportional Feedback illustration. Depending on the position of the user with respect to the track, the sound source is moved out of the track, in the opposite direction, by a distance Ke .

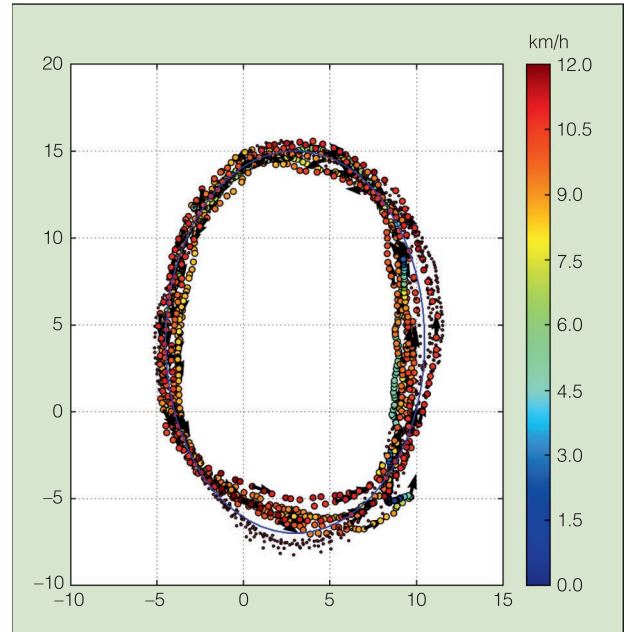


Fig. 7. Blind user practicing roller skating with a proportional-integral feedback with $K_p = 0.7$ and $K_i = 0.1$ (Blind subject #2).

Table 4 – Comparisons of measured error on elliptical courses for various users with and without feedback

User	Activity	Average error w/o PI	Average error with PI
Blind user #1	Roller skating	0.66 m	0.50 m
Blind user #2	Walking	0.37 m	0.16 m
Sighted #1	Brisk walking	0.80 m	0.70 m
Sighted #2	Walking	0.88 m	0.56 m

results obtained by different users with and without feedback and shows the improvement achieved by this method.

These guiding methods are effective. After a few low-speed laps, users (including blind people) are able to perceive the configuration of the route and can gain speed. The guidance system then allows the user to adjust his position in real time without any additional effort.

Conclusion

With the device described in this paper, we have been able to guide different people, including blind people, walking, running or roller skating in a protected environment, using spatialized sound stimuli. We have shown how to address the different issues of the problem in order to develop a real-time embedded system that contains HRTF filtering, depending on the location and head orientation of the user. We also provide the reader with results that show the efficiency of the approach.

Our experiments show that the user's reception of the device is very positive, and with very short training, the user follows the sound naturally and quite comfortably. Our prototype is regularly tested with users and athletes that help us to improve the system.

Much remains indeed to be done concerning the integration of the system in a better wearable device, but we believe that the concept developed in this article has been validated. In the long term, our system could help a greater number of visually impaired people to play sport, for example, with a single guide for an entire group. It could also facilitate the practice of competition.

Acknowledgments

We work in close partnership with several sport associations that help people with disabilities. We would like to acknowledge the associations "MCV," "Accessi Meca Sports" and all of the blind people and athletes who have contributed to these developments.

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Sylvain Ferrand (sylvain.ferrand@polytechnique.edu) is a Research Engineer and Ph.D. candidate at the Center for Applied Mathematics at the Ecole Polytechnique (CMAP), Palaiseau, France. After earning a master's degree in computer science, he is currently Head of IT and Scientific Computing Infrastructure at CMAP. His Ph.D. degree work mainly focus on audio 3D and efficient indoor localization for guiding people.

François Alouges (francois.alouges@polytechnique.edu) is Professor with the Department of Mathematics at Ecole Polytechnique in Palaiseau, France after previously obtaining a Full Professor position at the University of Orsay, France. He received his Ph.D. degree in 1990 and was a Researcher at the Centre National de la Recherche Scientifique in the ENS of Cachan, France. His interests mainly focus on numerical analysis and scientific computing for various problems such as 3D sound, acoustic simulation, continuum mechanics and low Reynolds number swimming.

Matthieu Aussal (matthieu.aussal@polytechnique.edu) is a Research Engineer at the Center for Applied Mathematics at the Ecole Polytechnique (CMAP), Palaiseau, France. After earning a master's degree in mathematical engineering at Paris Sud University, he defended an industrial thesis at CMAP, awarded by the agency Math-Industry (AMIES). As a specialist in numerical acoustics, his research focuses on 3D audio, auralization and waves equations resolution. He develops several software libraries with academic and industrial collaborations.